

1. Complete the answer for each statement. [4]

Statement	Answer
The hexadecimal value 2B represented in denary	
The denary number 200 represented in hexadecimal	
The denary number 45 represented in Binary Coded Decimal (BCD)	
The binary number 1011 1101 represented in hexadecimal	

2. Convert the hexadecimal number **A3C** into denary. Show your working. [2]

Working:

Answer:

3. A 16-bit register holds **0111 0100 1101 1001**. Give its value in hexadecimal. [2]

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4. (a) Convert the denary number **307** into BCD. [1]

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(b) Explain why the bit pattern **1010** can never appear as a nibble of a BCD number. [1]

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5. Describe **one** practical application of BCD and **one** of hexadecimal. [2]

BCD:

Hex:

6. A student claims that 58 in BCD and 58 in pure binary are the same bit pattern. Write both patterns and state the difference. [2]

BCD: Binary:

The Ice-Cream Challenge (0 marks. 1 ice cream.)



Write a program that ① reads a **binary string** of any length and prints it in hex, and ② reads a **hex string** and prints it in binary.

House rules: `int()`, `hex()` and `bin()` are banned — the machine may not do your homework. Loops, ifs and the 16-row nibble table are all you need. First working demo of **both directions** eats.

☛ **Want to dig?** BCD looks wasteful — question 6 showed you it needs more bits than plain binary for the same number. So why does it still exist? Find out what a **digital clock**, a **calculator display** or a **supermarket till** does with it, and why a machine that shows numbers to humans might rather store 58 as "5 then 8" than as 00111010. The answer is about what happens at the *edge* of the machine, not the middle of it.

draft / working